Homework 1
The improved SIR model

Michele Arcuri

Ottobre 2024

Introduction

During the course of Complex Systems, we studied the SIR model for the evolution of a pandemic in a population. In this work I propose the study of this same model, considering an interaction that brings this model closer to reality.

First we will see the differences between the mathematical definition of the Sir model and the improved SIR model, then we will find the equilibrium point for the improved model. We will then find the a formula to find R_n (the number of removed) which is also independent on the number of susceptive and infected. The main part of this work is simulating and study the system using different values of the parameters and different initial conditions.

1 The SIR model and the improved SIR model

This model is summarized in the system of equations on the left, where S_n represents the number of susceptible people (i.e., who may become ill), I_n represents the number of infected, and R_n represents the number of removed (i.e., the deceased and hospitalized). Furthermore, we impose $S_n + I_n + R_n = 1$ for each n.

$$\begin{cases} S_{n+1} = S_n - \mu S_n I_n \\ I_{n+1} = I_n + \mu S_n I_n - \gamma I_n \\ R_{n+1} = R_n + \gamma I_n \end{cases} \Rightarrow \begin{cases} S_{n+1} = S_n - \mu S_n I_n + \delta I_n \\ I_{n+1} = I_n + \mu S_n I_n - \gamma I_n \\ R_{n+1} = R_n + \gamma I_n - \delta I_n \end{cases}$$

On the right, instead, we see the system that we are going to study. We have added two terms that allow the removed elements to become susceptible again, in proportion to the number of infected.

We note how, in order to avoid getting a number of removed less than zero, we need to impose $\gamma \geq \delta$ (just imagine a situation with $R_0 = 0$ and $I_0 > 0$ and calculate the first iteration).

2 Equilibrium points

We need to solve

$$\begin{cases} S_n^* = S_n^* - \mu S_n^* I_n^* + \delta I_n^* \\ I_n^* = I_n^* + \mu S_n^* I_n^* - \gamma I_n^* \\ R_n^* = R_n^* + \gamma I_n^* - \delta I_n^* \end{cases} \Rightarrow \begin{cases} 0 = -\mu S_n^* I_n^* + \delta I_n^* \\ 0 = \mu S_n^* I_n^* - \gamma I_n^* \\ 0 = \gamma I_n^* - \delta I_n^* \end{cases}$$

We note that by substituting the third equation into the second, we obtain the first equation changed in sign. We can therefore state that the system is equivalent to

$$(\mu S_n^* - \gamma) I_n^* = 0, \quad \delta = \gamma$$

The stability condition is found in the same cases as in the SIR model, with the additive condition of $\delta = \gamma$.

3 The formula for R_n with no dependencies on S_n and I_n

We start from our system and rewrite it by highlighting the differences $S_{n+1} - S_n$ and $R_{n+1} - R_n$

$$\begin{cases} S_{n+1} &= S_n - \mu S_n I_n + \delta I_n \\ I_{n+1} &= I_n + \mu S_n I_n - \gamma I_n \\ R_{n+1} &= R_n + \gamma I_n - \delta I_n \end{cases} \Rightarrow \begin{cases} S_{n+1} - S_n &= (-\mu S_n + \delta) I_n \\ I_{n+1} &= I_n + \mu S_n I_n - \gamma I_n \\ R_{n+1} - R_n &= (\gamma - \delta) I_n \end{cases}$$

We now substitute I_n obtained from the third equation into the first one

$$S_{n+1} - S_n = (-\mu S_n + \delta) \cdot \frac{R_{n+1} - R_n}{\gamma - \delta}$$

$$\Rightarrow \frac{S_{n+1} - S_n}{-\mu S_n + \delta} = \frac{R_{n+1} - R_n}{\gamma - \delta}$$

$$\Rightarrow \frac{\mu S_{n+1} - \mu S_n}{-\mu S_n + \delta} = \mu \cdot \frac{R_{n+1} - R_n}{\gamma - \delta}$$

$$\Rightarrow \frac{\mu S_{n+1} - \mu S_n \pm \delta}{-\mu S_n + \delta} = \mu \cdot \frac{R_{n+1} - R_n}{\gamma - \delta}$$

$$\Rightarrow \frac{\mu S_{n+1} - \delta}{-\mu S_n + \delta} + 1 = \mu \cdot \frac{R_{n+1} - R_n}{\gamma - \delta}$$

$$\Rightarrow \mu S_{n+1} - \delta = (-\mu S_n + \delta) \left[\mu \cdot \frac{R_{n+1} - R_n}{\gamma - \delta} - 1 \right]$$

$$\Rightarrow \mu S_{n+1} - \delta = (\mu S_n - \delta)\left[1 - \frac{\mu}{\gamma - \delta} \cdot (R_{n+1} - R_n)\right]$$

$$\Rightarrow (\mu S_{n+1} - \delta) = (\mu S_0 - \delta)\prod_{i=1}^{n}\left[1 - \frac{\mu}{\gamma - \delta} \cdot (R_i - R_{i-1})\right] \qquad (1)$$

Now, because of the δ factors, it is not possible for us to iterate the procedure and obtain a product only for the quantity S_n.

Now we can take advantage of the fact that $1 = S_n + I_n + R_n$ and from

$$R_{n+1} - R_n = (\gamma - \delta)I_n$$

we get

$$R_{n+1} - R_n = (\gamma - \delta)(1 - S_n - R_n)$$

e

$$R_{n+1} - R_n = (\gamma - \delta)\left\{1 - \frac{1}{\mu}[(\mu S_n - \delta) + \delta] - R_n\right\}$$

substituting the formula that we just found for

$$R_{n+1} - R_n = (\gamma - \delta)\left\{1 - \frac{1}{\mu}\left[(\mu S_0 - \delta)\prod_{i=1}^{n}\left(1 - \frac{\mu}{\gamma - \delta} \cdot (R_i - R_{i-1})\right) + \delta\right] - R_n\right\}$$

and

$$R_{n+1} = (1 - \gamma + \delta)R_n + (\gamma - \delta)\left\{1 - \frac{(\mu S_0 - \delta)}{\mu}\prod_{i=1}^{n}\left(1 - \frac{\mu}{\gamma - \delta} \cdot (R_i - R_{i-1})\right) + \frac{\delta}{\mu}\right\}$$

At last, we obtained the equation of R_{n+1} that depends only on R_i with $i \leq n$.

4 First case: $\mu = 0.25$, $\gamma = 0.7$

To study the improved SIR model, we also made use of numerical simulations performed by the computer. In particular, we implemented a code in `c++` that would run the time progression of the system, we then used a python script to generate the graphs that we will see below (all the code is presented in the following link https://github.com/MIK2292/Modello_SIR.git).

We now impose the initial conditions of Italy (59 999 999 susceptible people and 1 infected), furthermore we choose $\mu = 0.25$, $\gamma = 0.7$ (the same parameters used in *APPUNTI 01 - Sistemi discreti e caos deterministico* given in Figure 1).

We can see how the similarity of the first graph in the upper left corner of Figure 2 with Figure 1 indicates the correctness of the executed code.

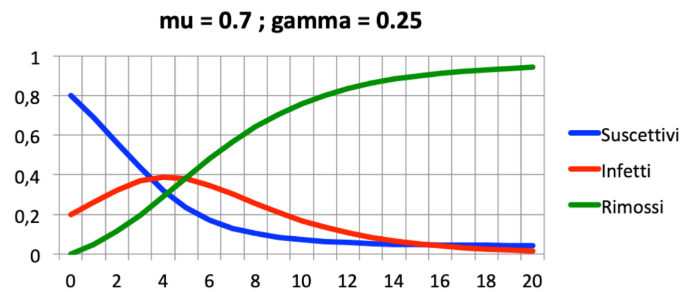

Figura 1: Example of evolution in time for the standard SIR model. Source: *APPUNTI 01 - Sistemi discreti e caos deterministico*

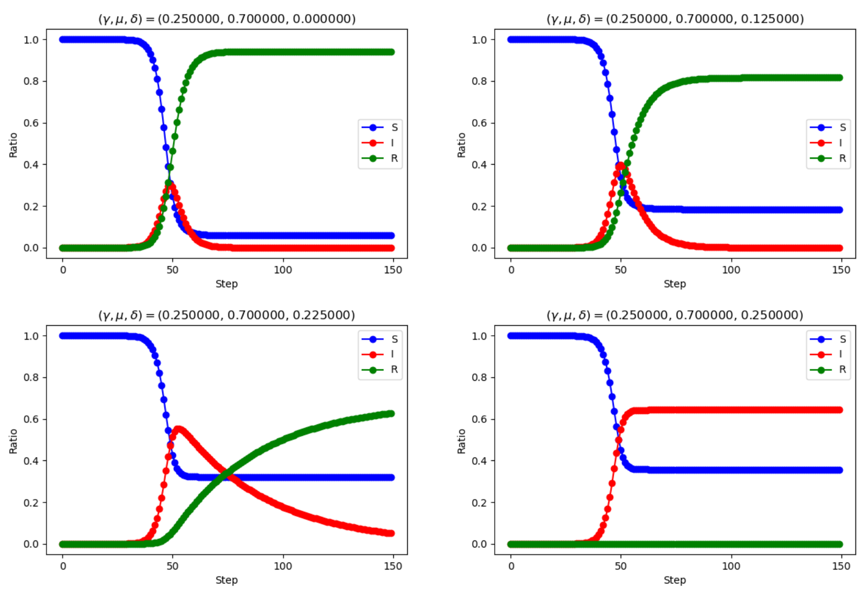

Figura 2: The time evolution of the improved SIR model, beginning from the initial conditions of Italy (59 999 999 susceptible people and 1 infected).

We can also see how in the lower right graph ($\delta = \gamma$) we find the stable condition studied above. In the intermediate cases, however, we see how the transition from one case to the other takes place.

Furthermore, Figures (3) and (4) show respectively the (I_{n+1}, I_n) space and the (S_n, I_n) space.

It's interesting to notice, in Figure (3), that the change in the value of δ pushes further from the origin the equilibrium point. This means that the value in the equilibrium point for the quantity I_n follows a direct proportionality with

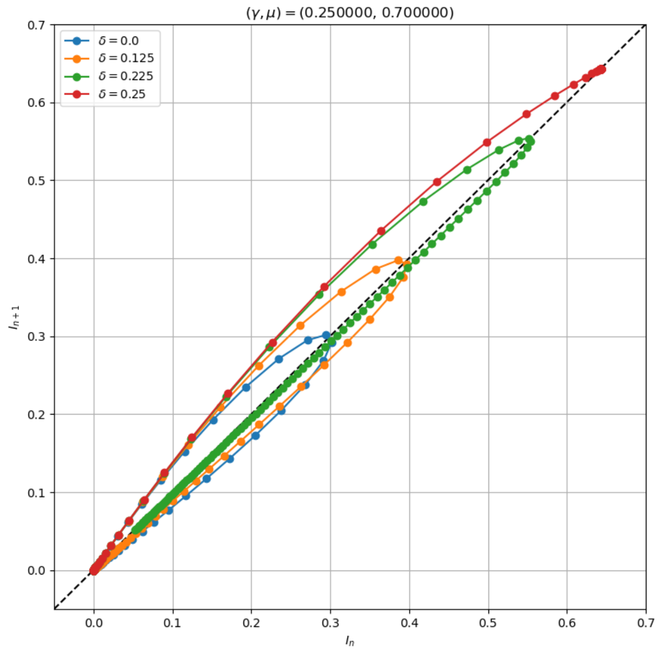

Figura 3: Space (I_{n+1}, I_n). Here, the points on the dashed bisection for each curve represent the equilibrium points.

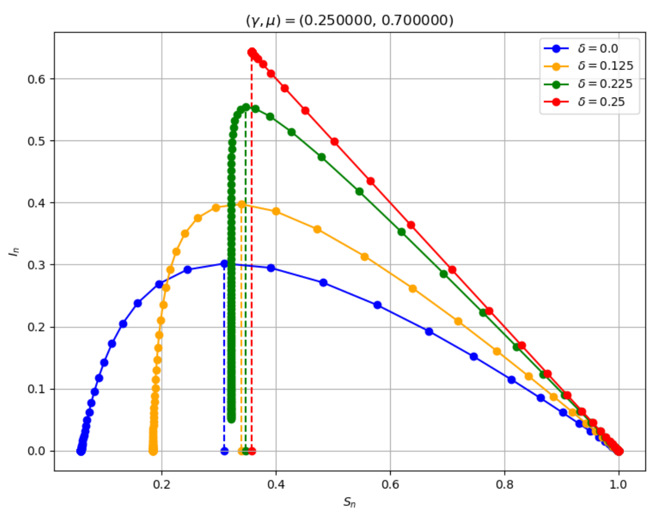

Figura 4: Space (S_n, I_n). Here, the point of maximum I_n for each curve represents the equilibrium point.

respect to δ. We also notice how for $\delta = 0.25 = \gamma$ we reach the equilibrium and the system remains stable in the equilibrium configuration, as we expected from the theory.

Figure (4) gives us insights on the quantity S_n. We see that its value for the equilibrium value also grows with δ.

5 Second case: $\mu = 0.5$, $\gamma = 0.45$

To verify the correctness of the model, we set $\delta = 0$ and run the simulation from the same initial conditions as the graphs found in the course notes (*APPUNTI 01 - Sistemi discreti e caos deterministico*).

Their similarity again indicates the correctness of the executed code.

Let's see what happens in the initial conditions of Italy (59 999 999 susceptible people and 1 infected). We change the value of δ in the range $0 \leq \delta \leq \gamma$ and observe the performance of the system in the following graphs (Figure 5) (note that the following graphs begin from step $x = 200$).

Conclusions

In this work we studied the SIR model using both analytical and and computational methods. We developed the code needed for the simulation, which can be easily reused for the study of other type of similar systems. At last, we found that the results given by the two methodologies lead to an agreement.

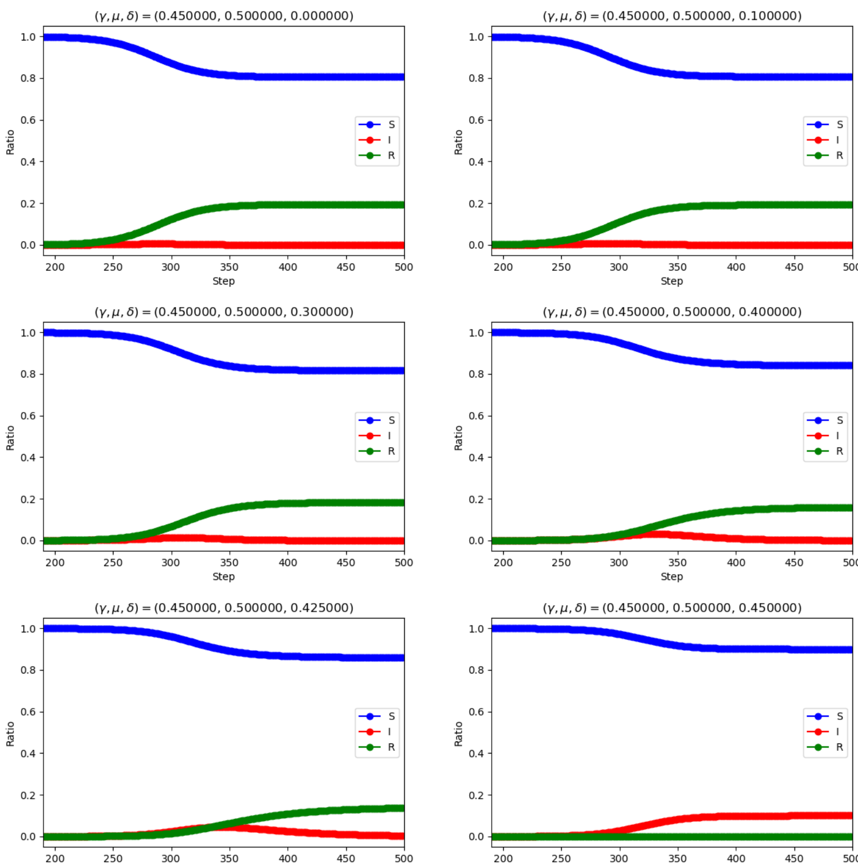

Figura 5: Evolution of the improved SIR model for different values for the parameter δ

References

1. *APPUNTI 01 - Sistemi discreti e caos deterministico*, Carbone V.

2. *Stochastic Dynamics and Irreversibility*, Tânia Tomé, Mário J. de Oliveira, 2015, Springer International Publishing